Job scheduling

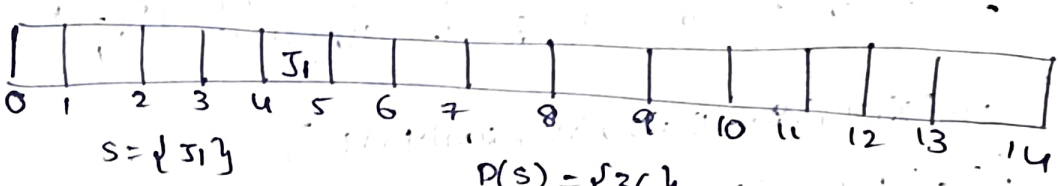
1) let there be 14 jobs for sequencing. whose profits are 36, 21, 28, 19, 21, 13, 28, 25, 18, 20, 27, 22, 14, 26.
Times are 5, 8, 3, 4, 4, 9, 12, 14, 2, 7, 5, 1, 6, 3

Sol

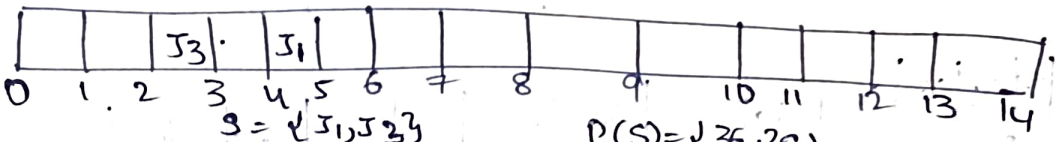
Arranging in descending order of profits;

P {36, 28, 28, 27, 26, 25, 22, 21, 21, 20, 19, 18, 14, 13}

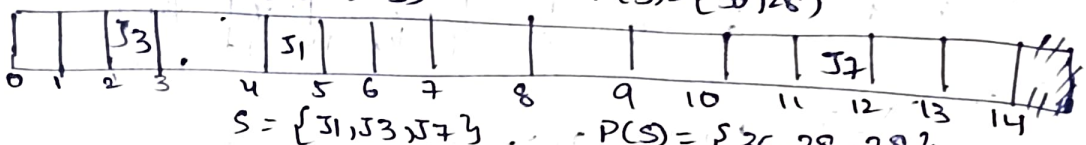
T {5, 3, 12, 5, 3, 14, 1, 8, 4, 7, 4, 2, 6, 9}



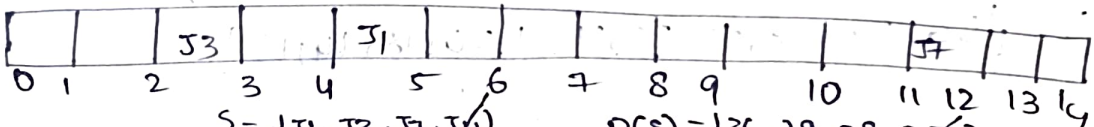
S = {J1} P(S) = {36}



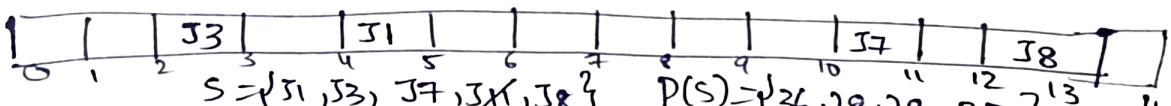
S = {J1, J3} P(S) = {36, 28}



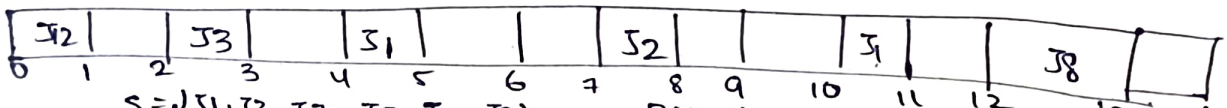
S = {J1, J3, J7} P(S) = {36, 28, 28}



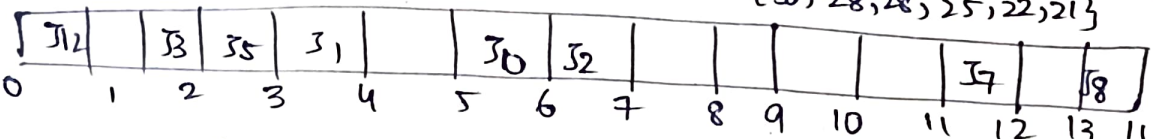
S = {J1, J3, J7, ~~J11~~} P(S) = {36, 28, 28, ~~27~~}



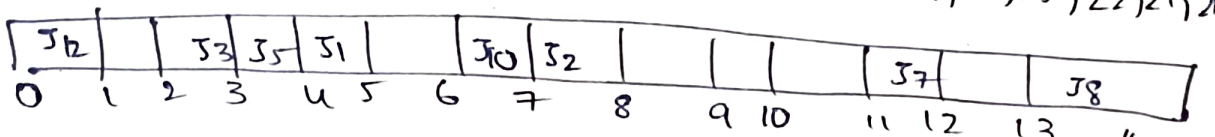
S = {J1, J3, J7, ~~J11~~, J8} P(S) = {36, 28, 28, 25}



S = {J1, J3, J7, J8, J12, J2} P(S) = {36, 28, 28, 25, 22, 21}



S = {J1, J3, J7, J8, J12, J2, J5} P(S) = {36, 28, 28, 25, 22, 21, 21}



S = {J1, J3, J7, J8, J12, J2, J5, J10} P(S) = {36, 28, 28, 25, 22, 21, 21, 20}

J12	J9	J3	J5	J1		J10	J2				J7		J8	
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14

S = {J1, J3, J7, J8, J12, J2, J5, J10, J9}

P(S) = {36, 28, 28, 25, 22, 21, 21, 20, 18}

J12	J9	J3	J5	J1	J13	J10	J2				J7		J8	
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14

S = {J1, J3, J7, J8, J10, J2, J5, J10, J9, J13}

P(S) = {36, 28, 28, 25, 22, 21, 21, 20, 18, 14}

J12	J9	J3	J5	J1	J3	J10	J2	J6				J7	J8	
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14

S = {J1, J3, J7, J8, J12, J2, J5, J10, J9, J13, J6}

P(S) = {36, 28, 28, 25, 22, 21, 21, 20, 18, 14, 13}

2Q) using Huffman coding, arrive max encoding bits for the phrase.

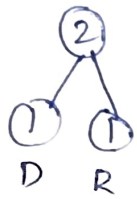
AMRITA VISHWA VIDYAPEETHAM CHENNAI CAMPUS

A	M	R	I	T	V	S	H	W	D	Y	P	E	C	N	U
7	3	1	4	2	2	2	3	1	1	1	2	3	2	2	1

Now, sort in Ascending order.

1	1	1	1	1	2	2	2	2	2	2	3	3	3	4
D	R	U	W	Y	C	N	P	S	T	V	E	H	M	I

→



1	1	1	2	2	2	2	2	2	3	3	3	4
U	W	Y	C	N	P	S	T	V	E	H	M	I

→

2	2	2	2	2	2	3	3	3	4
C	N	P	S	T	V	E	H	M	I

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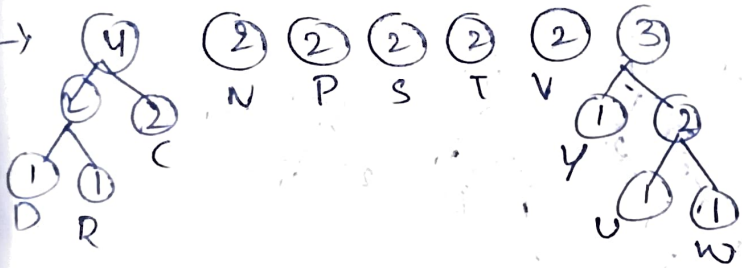
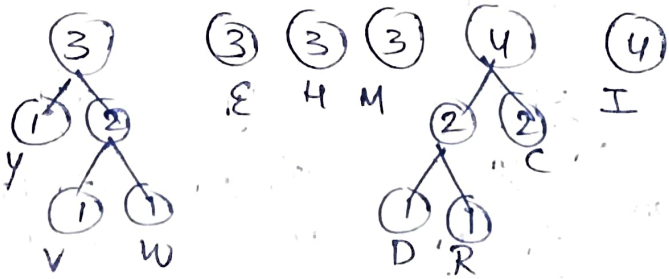
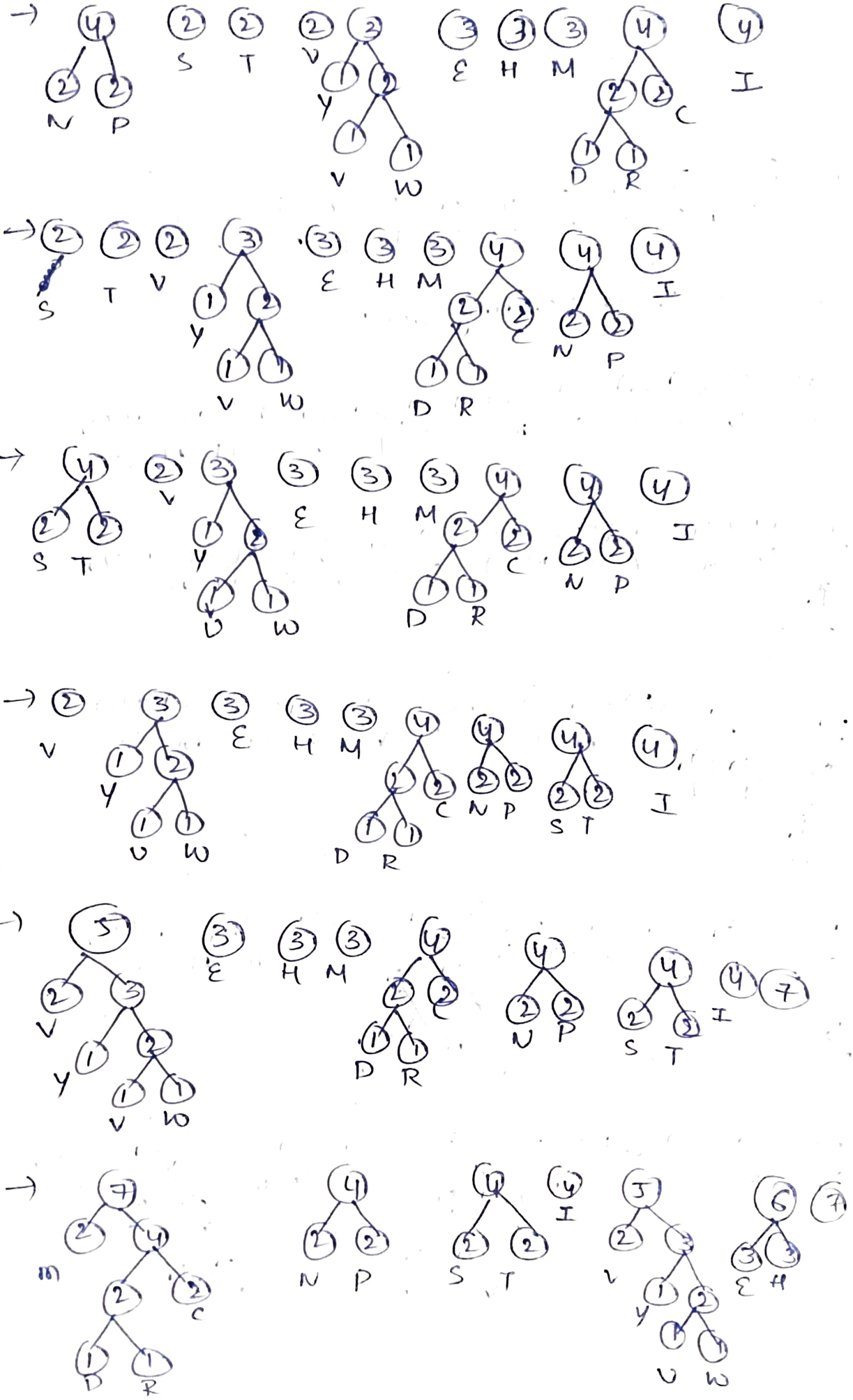


2	2	2	2	2	3	3	3	3	4
N	P	S	T	V		E	H	M	I

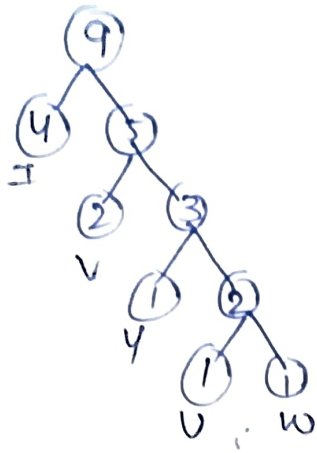
→



2	2	2	2	2	3	3	3	3	4	4
N	P	S	T	V		E	H	M		I


N P S T V Y V W E H M C D R I
S T V Y V W E H M C D R N P I
S T V Y V W E H M C D R N P
V Y V W E H M C D R N P S T I
V Y V W E H M C D R N P S T I
m D R C N P S T I V Y V W E H

→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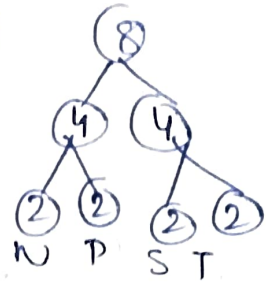
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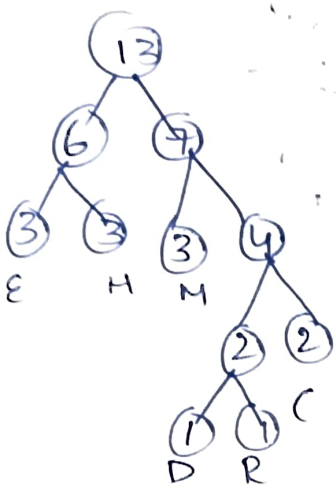
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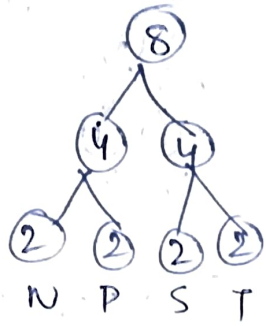
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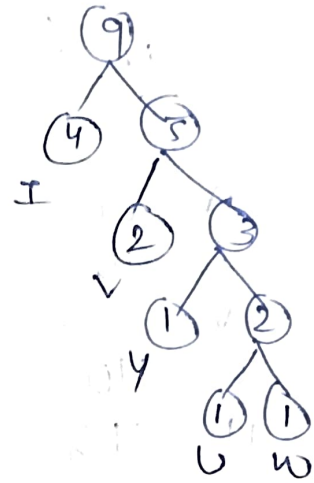
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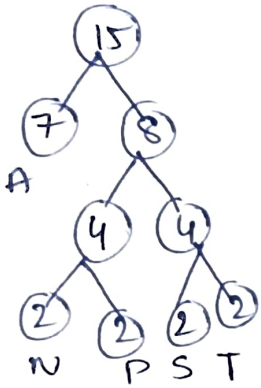
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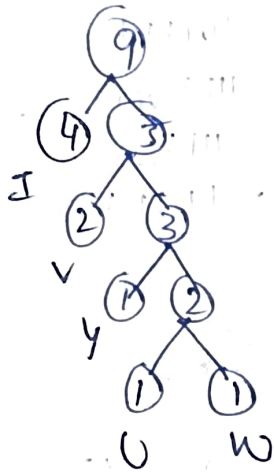
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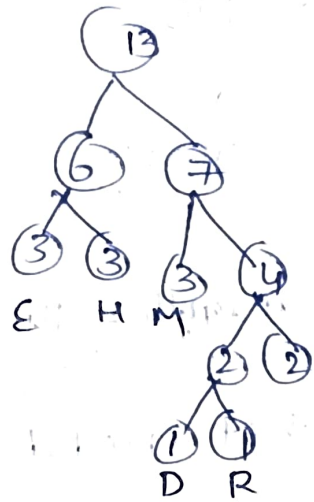
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R

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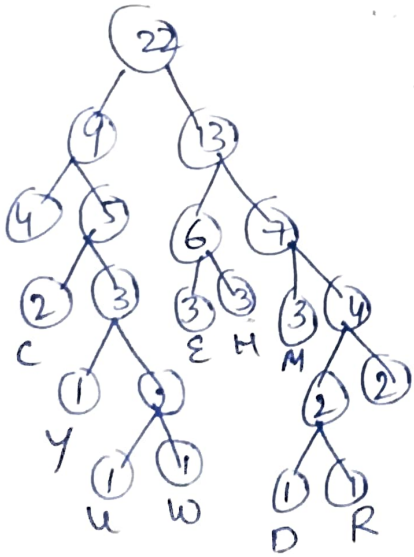
22
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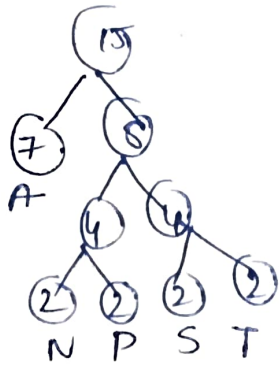
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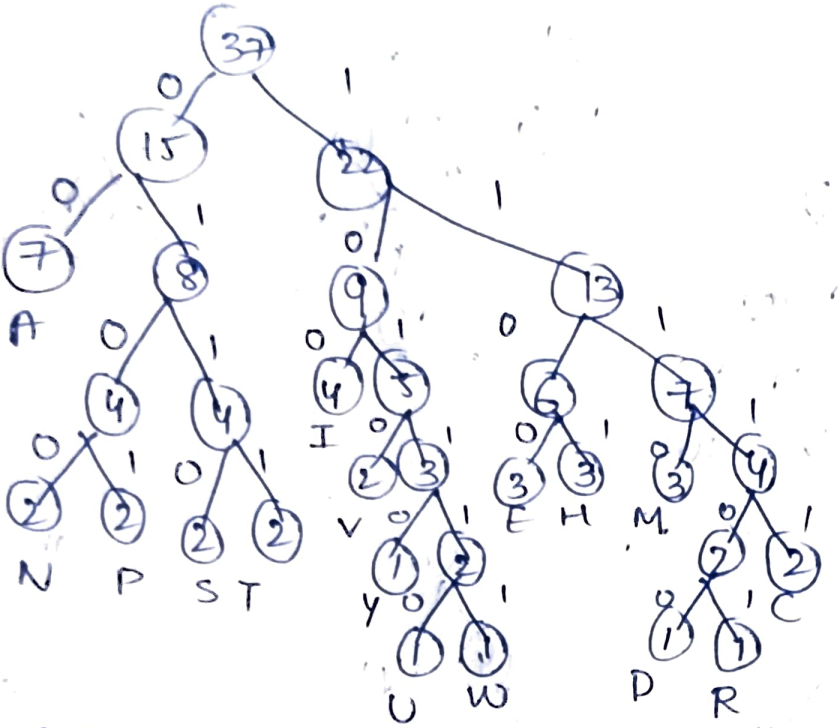

A - 00
I - 100
V - 1010
E - 1100
H - 1100
E - 1101
M - 1110

N - 0100
P - 0101

S - 0110
T - 0111
Y - 10110
U - 101110
W - 101111
D - 111100
R - 111101
C - 11111

Avg code length $= \frac{141}{37} = 3.81$

Length of encoded msg $\Rightarrow 37 \times 3.81$
$= 141.$